Cost analysis — PixelPine rev A

Planning rate **1 USD = 0.9 EUR**. Generated by `cost_model.py` — change an input there and re-run instead of editing numbers here. Items marked *EST* are engineering estimates; everything else is a list price (LCSC, JLCPCB published PCBA fees, fetched 2026-09-23).

Assembly volume per board: **552 SMD joints** (480 of them are LEDs), 0 THT joints, one side only.

Unit cost by volume

	Pilot · 100 pcs	Batch · 1 000 pcs	Volume · 10 000 pcs
How	JLCPCB Economic PCBA, parts from JLC/LCSC stock	JLCPCB Standard PCBA, panelised 2-up	Shenzhen contract manufacturer, LEDs bought direct
Components (BOM)	$1.75	$1.57	$1.14
Bare PCB incl. stand fin *(EST)*	$0.950	$0.400	$0.220
SMT assembly (joints + amortised setup)	$1.07	$0.717	$0.567
Programming (SWD pogo jig)	$0.0000	$0.090	$0.020
Stand fitting, test, packing labour	$0.0000	$0.190	$0.040
Packaging	$0.030	$0.180	$0.080
Ex-works cost	**$3.80**	**$3.15**	**$2.06**
Freight to CZ	$0.300	$0.240	$0.050
Landed in CZ (+3.7% duty, broker)	**$4.25 / €3.83**	**$3.55 / €3.20**	**$2.20 / €1.98**
Compliance amortised over the batch	$30.00	$3.00	$0.300
Total per unit	**$34.25 / €30.83**	**$6.55 / €5.90**	**$2.50 / €2.25**

Compliance budget (first year, one-off): EMC test at an accredited lab (EN IEC 55015 + EN 61547) €2,000, RoHS evidence (supplier declarations + 1 XRF screen) €400, WEEE + packaging EPR registration, first-year fees (CZ) €300 → **€2,700**. Details in eu-compliance/.

From cost to shelf price

Retail prices include 21 % Czech VAT. Retail channel assumes the usual keystone (retailer takes ~50 % of the net price); own e-shop assumes ~12 % payment/platform fees and a €0.60 shipping subsidy.

Channel	Shelf price	Net of VAT	Our revenue / unit	Margin @1k	Margin @10k
Retail / gift shop	€12.90	€10.66	€5.33	−€0.57	€3.08
Own e-shop, Etsy, Tindie	€14.90	€12.31	€10.24	€4.34	€7.99
B2B promo (custom silkscreen logo), 500+ MOQ	quoted ex-VAT	€6.90	€6.90	€1.00	€4.65

What the numbers say

- **The LEDs drive everything.** At 1 000 pcs the LEDs are 86% of the BOM, and their 480 joints are 87% of the assembly bill. Halving the LED count would cut the unit cost by roughly a third — but 240 dense LEDs *are* the product; keep them.

- **The MCU is almost free.** The PY32F003F18P6 (64 KB flash) is ~$0.13 in reels, cheaper than the 16 KB CH32V003 it replaced; a charlieplex driver chip such as the IS31FL3731 on the reference board costs over 10× more.
- **At 100 pcs compliance kills the margin** — treat it as a pilot (samples, school fair, beta testers) and keep those units off the market until the EMC report exists. At 1 000 pcs the classic retail channel loses money because the €2 700 compliance budget is spread over too few units; own e-shop and B2B promo orders carry it. From ~3 000 pcs retail works, and at 10 000 pcs every channel is healthy.
- **Competing with AliExpress** (bare soldering kits at $2.5–4.3 shipped) on price alone is not possible at the retail shelf — the landed volume cost is close to their selling price, and EU compliance plus VAT is a cost those listings often skip. The product has to win on looks (dense warm-white matrix, animations, ready-built) and sit in the €10–15 gift slot. See market-analysis.md.

Cost-down levers not yet taken

Lever	Saving / unit	Trade-off
Direct LED reel quote instead of LCSC list	≈ $0.40	needs 250 k+ LEDs per order
Cold-white basic LED (C2290)	≈ $0.05 + $3/order	loses the warm look
Drop C1 (10 µF) and rely on C2 + USB source capacitance	≈ $0.02	more supply ripple on touch sensing
Stand fin from the PCB panel's waste corners instead of its own area	≈ $0.03	tighter panel layout
Kraft paper sleeve instead of a printed box	≈ $0.10	less gift appeal